自动聚类模型

架构说明

这套方案是一个分流式聚类架构，由 3 个存储过程组成：

- sp_embedding_clustering_pipeline_router（总入口）
- sp_embedding_cluster_large_std（大样本链路）
- sp_embedding_cluster_llm_small_std（小样本链路）

核心思想：

- 先统计输入数据量
- 超过阈值走“大样本链路”（embedding + K-means）
- 不超过阈值走“小样本链路”（直接 LLM 语义聚类）
- 两条链路最终写入同一 cluster 结果表，便于统一分析

参数含义

sp_embedding_clustering_pipeline_router 参数：

- _query：输入数据 SQL，必须产出 id/content/callout_id/date
- _embedding_prompt：单条会话摘要提示词（大样本链路使用）
- _cluster_table_name：聚类结果表名（大小样本共用）
- _start_date：聚类统计起始日期（大样本链路）
- _end_date：聚类统计结束日期（大样本链路）
- _k：K-means 聚类数（大样本链路）
- _large_cluster_prompt：大样本簇解释提示词
- _sample_threshold：分流阈值（行数）
- _small_top_n：小样本输出 topN 类别数
- _small_cluster_prompt：小样本直接 LLM 聚类提示词

如何输入

推荐在一个脚本里先 declare 三段文本参数（_query、_embedding_prompt、_large/_small_cluster_prompt），再 call 总入口。_query 是唯一硬约束输入，列名必须对齐：

- id
- content
- callout_id
- date

输出表结构（统一口径）

无论走大样本链路还是小样本链路，最终都写入同一结构的结果表（_cluster_table_name）：

- callout_id STRING：分组标识
- clusters STRING：簇编号/类别编号
- cluster_result STRING：簇的一句话总结（最终展示字段）
- id STRING：会话唯一标识
- llm_result STRING：会话摘要文本（用于聚类输入/回溯）
- date DATE：会话日期
- distance_to_cluster FLOAT64：样本到簇中心距离（仅大样本 K-means 有值，小样本为 NULL）
- ml_summary STRING：近邻样本解释（仅大样本有值，小样本为 NULL）
- llm_summary STRING：簇级 LLM 解释

输入示例

代码块

```
declare _query string default
'''
select distinct
  cast(a.sessionid as string) as id,
  b.msg_dialogue as content,
  'ja_all' as callout_id,
  date(a.createtime) as date
from `trip-ibu-ser-dbprj.ibu_ser_dw_cdw.nova_ser_ib_chat_ai_analysis` a
left join `trip-ibu-ser-dbprj.ibu_ser_dw_cdw.mid_ser_ib_chatmsg_agg` b
  on a.sessionid = b.sessionid
where a.effective = '1'
  and a.biztype = '1322'
  and a.sessionmode in ('C2O', 'S2C')
```

```
  and lower(a.language) in ('ja')
  and a.createdate >= '2025-12-10'
  and a.createdate <= '2025-12-12'
  and a.scenario in ('Change 改签', 'Refund 退票', 'Flight Change 航变')
  and upper(a.sessiontype) = 'AGENT'
''';

declare _embedding_prompt string default
'''You are an OTA incident analyst.
Extract the root trigger event behind this contact.
Output one concise English sentence only.''';

declare _large_cluster_prompt string default
'''You are an OTA incident analyst.
Given multiple conversation summaries in one cluster, summarize the dominant concrete triggering event.
Output one concise English sentence.''';

declare _small_cluster_prompt string default
'''Cluster the input records into top categories and return JSON with clusters and assignments only.''';

call `trip-ibu-ser-dbprj.ibu_ser_dw_cdw.sp_embedding_clustering_pipeline_router`(
  _query,
  _embedding_prompt,
  'adm_prj_ssr_alert_msgdialogue_cluster',
  '2025-12-10',
  '2025-12-12',
  10,
  _large_cluster_prompt,
  500,
  20,
  _small_cluster_prompt
);
```

结果分析

```sql
with t_day_total as (
  select
    c.date,
    count(*) as day_total
  from `trip-ibu-ser-dbprj.ibu_ser_dw_cdw.adm_prj_ssr_alert_msgdialogue_cluster` c
  where c.date between date('2025-12-10') and date('2025-12-12')
  group by c.date
)
select
  c.callout_id,
  c.clusters as cluster,
  count(*) as total_cnt,
  countif(c.date = date('2025-12-10')) as cnt_d1,
  countif(c.date = date('2025-12-11')) as cnt_d2,
  countif(c.date = date('2025-12-12')) as cnt_d3,
  any_value(c.ml_summary) as ml_summary,
  any_value(c.llm_summary) as llm_summary
from `trip-ibu-ser-dbprj.ibu_ser_dw_cdw.adm_prj_ssr_alert_msgdialogue_cluster` c
left join t_day_total t
  on c.date = t.date
where c.date between date('2025-12-10') and date('2025-12-12')
group by c.callout_id, c.clusters
order by total_cnt desc
```

查询已完成

查询结果

+ 创建

作业信息 | 结果 | 可视化图表 | JSON | 执行详细信息 | 执行图

行	callout_id	cluster	total_cnt	cnt_d1	cnt_d2	cnt_d3	ml_summary	llm_summary
1	ja_all	8	137	42	48	47	Nearest samples: Trigger event: Airline operational disruption; Impact: The original flight was canceled by the airline, and the customer contacted support to understand the reason for the cancellation and the status of their pending change request. || Trigger event: airline operational disruption; Impact: flight cancellation and significant delays led the customer to contact support for a rebooking and	Widespread airline operational disruptions, including flight cancellations, schedule changes, and overbooking, have triggered a surge in customer requests for rebooking and fee clarifications.
2	ja_all	1	120	57	34	29	Nearest samples: Trigger event: airline operational disruption; Impact: the flight from Osaka to Kunming was cancelled by the airline, and the customer contacted support to request a free rebooking to an alternative flight. || Trigger event: airline operational disruption; Impact: the customer's flight from Osaka to Harbin was cancelled by the airline, and the customer contacted support to request a free	The dominant triggering event is the widespread cancellation of flights between Japan and China by the airline due to operational disruptions.
3	ja_all	3	98	34	39	25	Nearest samples: Trigger event: airline operational disruption; Impact: flight cancellation led the customer to request a full refund and assistance with rebooking a new itinerary. || Trigger event: airline operational disruption; Impact: flight schedule change and subsequent cancellation led the customer to seek a free rebooking or full refund. || Trigger event: airline operational disruption;	The dominant triggering event is widespread airline operational disruptions involving involuntary flight cancellations and significant schedule changes across multiple itineraries.
4	ja_all	6	90	37	24	29	Nearest samples: Trigger event: unknown; Impact: The customer contacted support to cancel an existing flight change request and submit a new one because their preferred return route (Barcelona-Beijing-Fukuoka) appeared sold out during the processing period. || Trigger event: unknown; Impact: The customer contacted support to change their flight from Fukuoka to Beijing after a previous	Customers contacted support primarily due to complications and inquiries surrounding flight change requests, refund eligibility, and ticket cancellations across various international routes.
5	ja_all	4	73	30	27	16	Nearest samples: Trigger event: Airline operational disruption; Impact: China Eastern Airlines flight schedule change due to plan adjustments led the customer to inquire about full refund eligibility and partial cancellation options. || Trigger event: Airline operational disruption; Impact: Air China flight cancellation led the customer to request reimbursement for hotel and train expenses and a formal	Multiple airlines, including Air China and China Eastern, implemented widespread flight cancellations and schedule changes on routes between China and Japan (notably Fukuoka, Nagoya, and Osaka).

Andy Zhang (张宗煜) TR058985 2026年04月07日 15:49